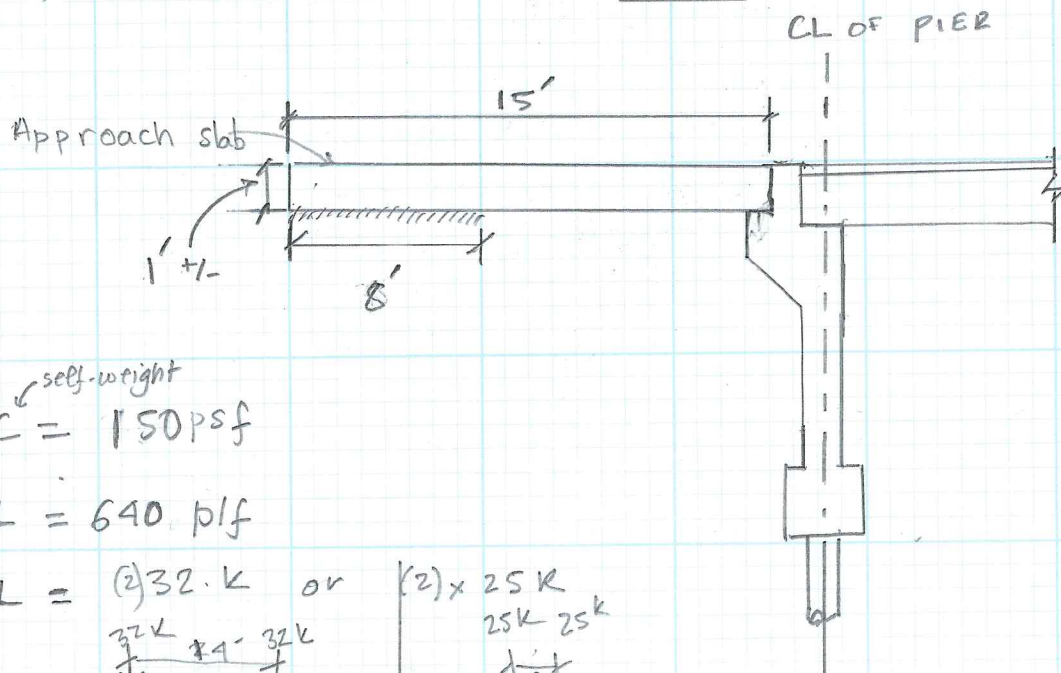


Approach slab loading

Total Length, $L = 15'$

Bearing length $= \frac{15}{3} = 5'$ or $\frac{25}{3} = \underline{8'}$

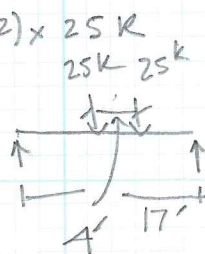
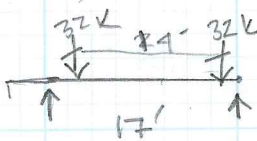
Effective Span length, $L_{eff} = 25 - 8 = \underline{17'}$



self-weight
DC = 150 psf

DL = 640 plf

LL = (2) 32 k or (2) x 25 k



Worst case
Reactions $R = 32 + 32 \times \frac{(17-14')}{17}$
 $= 37.6 k$

$V = 25 k \times 2 = 50 k$ governs

$50 k \times (1) \times 1.20 = 60 k$

$50 k \times (2) \times 0.85 = 85 k$

$50 k \times (3) \times 0.65 = 97.5 k$ governs

DL $R = 3.2 k$ $12 = 3.4 k$

DC $R = 1 k$ $R = 1 k$

Project Davis Bridge

Location

Client

By LM

Date 06/18/20

Revised

Date

Sheet No.

Job No.

Approach slab Loading



$$DL = 0.64 \text{ k/ft} + \frac{3.2 \text{ k}}{5 \text{ ft}} = 1.28 \text{ k/ft} \text{ 1 ft strip}$$

$$DC = 0.2 \text{ k/ft} + \frac{2.4 \text{ k}}{5} = 0.4 \text{ k/ft} \text{ 1 ft strip}$$

$$DL = 1.28 \text{ k/ft} \times 12 \times 3 = 46 \text{ k/ft}, 97 \text{ k} \leftarrow \text{Tandem}$$

$$DC = 0.4 \text{ k/ft} \times 43 = 17.2 \text{ k/ft}$$

$$LL = (46 \text{ k} + \frac{97 \text{ k}}{5}) \times 1.33 = 87 \text{ k/ft}$$

$$DL = 20 \text{ k/ft}$$

Project **DAVIS BRIDGE**

By **IM**

Sheet No.

Location

Date **6/19/20**

Client

Revised

Job No.

Date

Bridge width = 43'

Approach Length = 15'

Approach bearing length = $\frac{15'}{3} = 5'$

Approach span = $15' - 5' = 10'$

LL, $q = 87 \text{ k} / (43') = 2 \text{ k/ft/ft}$

DL, $q = 20 \text{ k} / (43') = 0.45 \text{ k/ft/ft}$

DL, $q = 0.45 \text{ k/ft}^2$

LL, $q = 2.00 \text{ k/ft}^2$

